Exam Date & Time: 21-Apr-2023 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (MAKEUP) EXAMINATIONS - APRIL 2023
SUBJECT: BIO-1071 / BIO -1051 - BIOLOGY FOR ENGINEERS
(REGULARS - LATE ADMISSION BATCH)

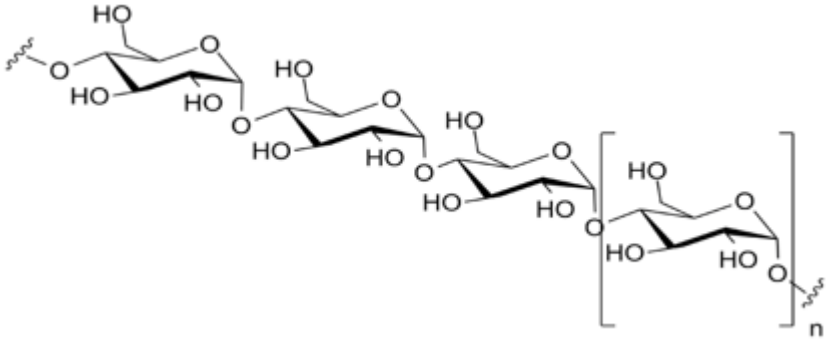
Biology for Engineers [BIO-1071]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

1A) (i) What type of the structure is given below?
(ii) What kind of Bond formation will occur between the molecules in the given example and how?



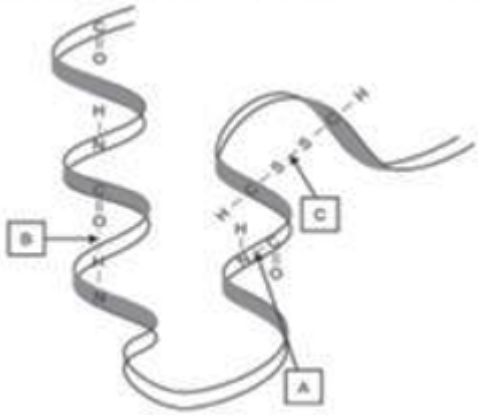
(3)

(1+2 = 3 marks)

1B) (i) There is a coating on the feathers of waterfowl such as ducks that keeps them from getting wet. What molecule would this water-repelling coating be made of?
(ii) How does enzymes affect enzyme catalysed reaction?
(iii) If your fever goes too high - over 105°F - what happens to the enzymes?
(1+1+1 = 3 marks)

(3)

1C) Enzymes, which are composed mostly of protein, catalyse many chemical reactions. The structure of a portion of an enzyme, with some of its constituent atoms shown, is represented below.
(i)Name the type of chemical bond present in the parts labelled as A, B, C.
(ii) Which chemical bond (A, B or C) is disrupted if the enzyme is denatured?



(4)

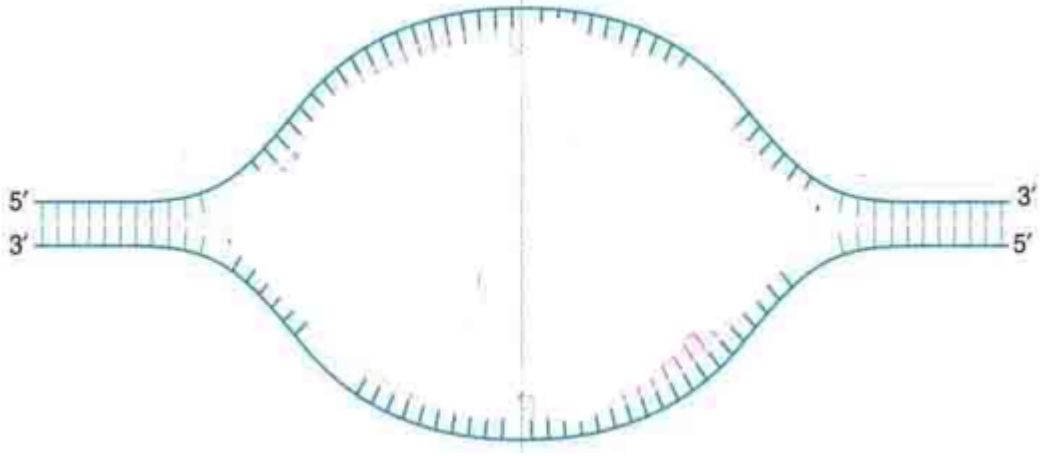
(3+1 = 4 marks)

2A) A colour-blind father has a daughter with normal vision. Mother had a normal vision. The daughter marries a man with a normal vision.
(i) What is the probability of her son and daughter to be colour blind? (3)
(ii) Explain with the help of a pedigree chart.
(1½+1½ = 3 marks)

2B) Mendel planned the experiment in such a way to analyze. Whether the alleles maintain the association they had in the parental generation:
For this he crossed pure breeding spherical seed and yellow seed color pea plant with a wrinkled seed green seed color pea plant. If the alleles maintain the association, he is expecting only the parental types in the F2 generations (Why not F1 generation)? (3)
(i) Construct punnet square and justify the parental types.
(ii) What kind of assortment is proved in the above problem?
(2+1 = 3 marks)

2C) i) The gene for hair color in rabbits has two alleles Q and q. Q is dominant and codes for brown hair. q is recessive and codes for white hair. Write out all the possible genotypes and phenotypes.
ii) Using the above example, fill in the Punnett's Square of offspring genotypes if one parent is heterozygous and the other is white haired. If the pair of rabbits have a litter of 24 babies, write out the expected number of each genotype and phenotype in the form of the table. (4)
(2+2 = 4 marks)

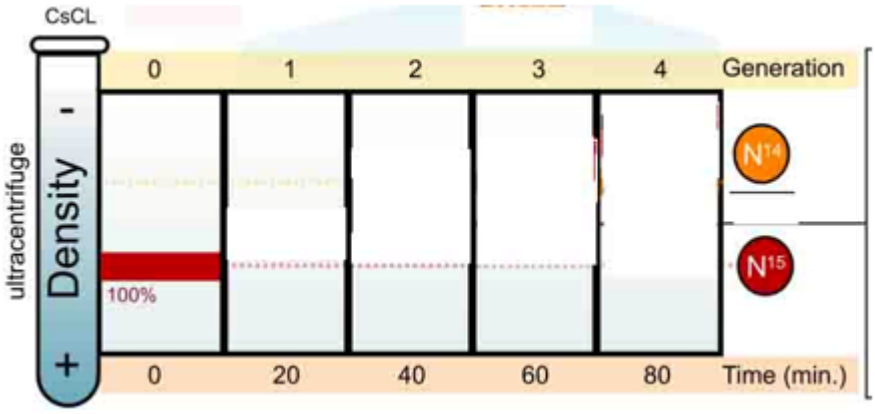
3A) Given below is a replication fork with origin of replication is in the middle. Complete the fork by constructing (i) Leading strands, Lagging strands (ii) Location of primers (iii) Ends of leading and lagging strands (iv) Location of DNA helicase enzyme (v) Location of DNA Polymerase enzyme (f)direction of fork movement.



(3)

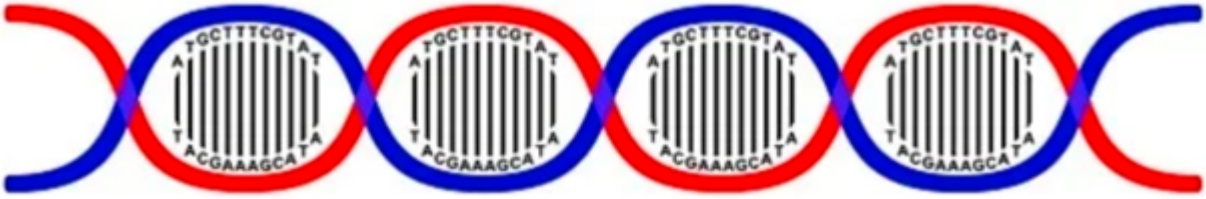
(½ mark × 6 = 3 marks)

3B) i) If an RNA has 28% adenine then what will be the percentage of guanine, is it possible to find? Justify your reasoning (3)
ii) A replication fork moves 1000 nucleotides per second, including leading and lagging strands. Calculate the number of origin of replications required to finish complete replication of a chromosome of size 201600000 bp in 40 minutes.
iii) Construct a diagram illustrating the relative position of bands in $CsCl_2$ centrifugation for dispersive mode of replication for in the following illustration.



(1+1+1 = 3 marks)

3C) Given below is a DNA sequence and the protein produced from it. Analyze the sequences and answer the following questions.



Protein produced from this DNA is = Methionine-Leucine-Serine-Tyrosine-Serine-STOP

i) Label the ends of the DNA.

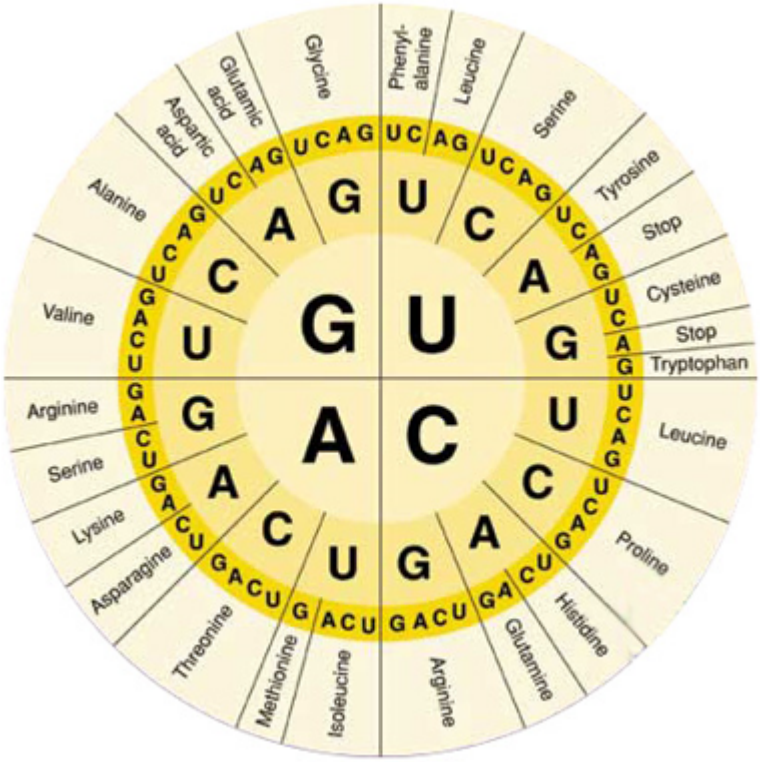
ii) Construct an appropriate stop codon for the above DNA.

iii) Construct the mRNA sequence that can synthesize the given protein.

iv) Label the complete non-coding region of the DNA.

v) Label the C terminus and N terminus of the protein.

(4)



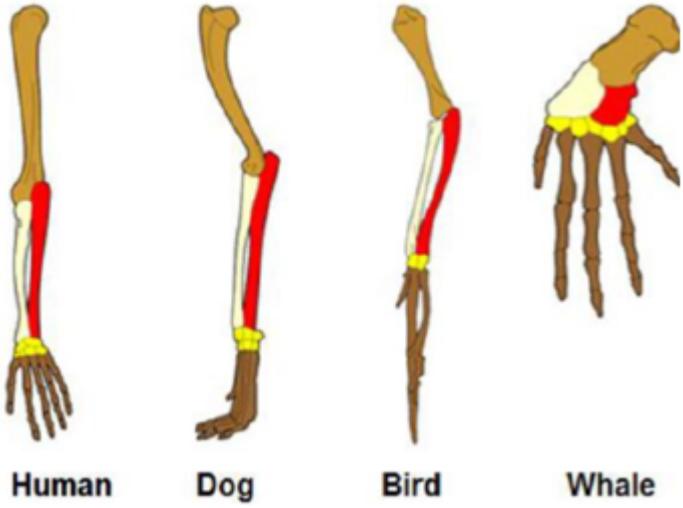
(½+½+2+½+½ = 4 marks)

4A) Explain the role of mutation in evolution. (3)

4B) What are Darwins logical findings? (3)

4C) (4)

Explain the idea of evolution illustrated in the following diagram:



5A) Rohit, a father of newly born child asks to the child specialist that weather with adequate levels of hygiene, sanitation and clean water, is there still a need for vaccination of my child? Will all the vaccines be 100% effective? Prepare answer with logical reasoning. (3)

5B) What is the logic that engineers should understand living systems for better and feasible innovation? Justify. (3)

5C) What is the concept for bioinspiration in the following cases? Explain

Example of source/product	Concept	Explanation

(4)

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